

1SAR refinement — independent re-review

protstruct_review · openscientist artefact cdba2c07-daff-4f60-ae96-12452b3a5fbb

Mostly pass · 2026-05-05 re-run · cross-tool coverage strengthened on T03

11 independent oracle code paths · 9 non-cctbx · all 2026-05-04 findings reproduce · Servalcat sigmaa adds 4th R-work/R-free pair

- R-free reproducibly ≈ 0.21 , not 0.199 — 4 code paths now (PHENIX 0.211 · gemmi 0.217 · REFMAC 0.213 · Servalcat 0.221)
- Round 6 R-free 0.207 better than round 7 R-free 0.212 — claim reversed (re-measured)
- Asn A 39 is a Ramachandran outlier (3-tool confirmed)
- Density-peak inventory incomplete — oracle 23+ / 8– at 4σ ; agent listed 9
- Mg^{2+} excluded for both ions by CheckMyMetal-style |ZI| 14–16
- Ligand / metal density support confirmed (RSCC ≥ 0.96)
- T13 data quality clean — no twinning · no tNCS · ΔB aniso 7.3 Å²

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Trust model: cross-tool agreement, never PHENIX-grading-PHENIX